

Lindblad equation for impurity model

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Total Hamiltonian

The total Hamiltonian of the system (impurity), bath (a fermion lattice) and their interaction is

$$H_T = H + H_B + \alpha H_I \quad (1)$$

where,

$$\begin{aligned} H &= \sum_{\sigma} E_0 f_{\sigma}^{\dagger} f_{\sigma} + U n_{+} n_{-} \\ H_B &= \sum_{k, \sigma} \epsilon_k c_{k, \sigma}^{\dagger} c_{k, \sigma} \\ H_I &= \sum_{k, \sigma} V_k f_{\sigma}^{\dagger} c_{k, \sigma} + H.c. \end{aligned} \quad (2)$$

Our aim is to derive the Lindblad equation to provide an approximate description of the system's dynamics under the Born-Markov approximation.

Assumptions of Lindblad master equation

There are three approximations in the microscopic derivation of the Lindblad master equation, resulting in some reasonable assumptions:

- ① **Born approximation:** Weak coupling between system and bath.
- ② Separable initial state: $\rho_T(0) = \rho(0) \otimes \rho_B(0)$
- ③ Bath in equilibrium: $[H_B, \rho_B(0)] = 0 \implies \rho_B(t) = \rho_B(0)$.
- ④ Influence of the system on the bath is small:
 $\rho_T(t) \approx \rho(t) \otimes \rho_B(t) = \rho(t) \otimes \rho_B(0)$
- ⑤ **Markov approximation:** "What happens next depends only on the state of affairs now." Underlying basic physical condition: reservoir (bath) correlation functions decay sufficiently fast.

(Redfield equation obtained with above assumptions)

- ⑥ **Rotating-wave approximation (RWA):** ensure complete positivity

Note: A decay of reservoir correlation can only be strictly valid for an infinitely large bath involving continuum frequencies.

Born approximation*

In interaction picture ($\hat{O}(t) \equiv e^{i(H+H_B)t} O(t) e^{-i(H+H_B)t}$), the dynamic equation of the total density matrix $\hat{\rho}_T(t)$ is

$$\begin{aligned} \frac{d\hat{\rho}_T}{dt} &= -i\alpha[\hat{H}_I(t), \hat{\rho}_T(t)] \\ \implies \hat{\rho}_T(t) &= \hat{\rho}_T(0) - i\alpha \int_0^t ds [\hat{H}_I(s), \hat{\rho}_T(s)]. \end{aligned} \quad (3)$$

Plugging the solution back into the dynamic equation, we have

$$\frac{d\hat{\rho}_T}{dt} = -i\alpha[\hat{H}_I(t), \hat{\rho}_T(0)] - \alpha^2 \int_0^t ds [\hat{H}_I(t), [\hat{H}_I(s), \hat{\rho}_T(s)]]. \quad (4)$$

The above equation is clearly non-local in time as the entire history $\hat{\rho}_T(s)$ is involved. However, as $\hat{\rho}_T(t_2) - \hat{\rho}_T(t_1) = -i\alpha \int_{t_1}^{t_2} dt' [\hat{H}_I(t'), \hat{\rho}_T(t')]$, replacing $\hat{\rho}_T(s)$ with $\hat{\rho}_T(t)$ in (4) changes the RHS by a term of $\mathcal{O}(\alpha^3)$. Omitting it is valid under Born approximation (weak coupling).

Born approximation*

The approximated time-local dynamic equation of the total density matrix under Born approximation is

$$\frac{d\hat{\rho}_T}{dt} = -i\alpha[\hat{H}_I(t), \hat{\rho}_T(0)] - \alpha^2 \int_0^t ds [\hat{H}_I(t), [\hat{H}_I(s), \hat{\rho}_T(t)]] \quad (5)$$

As we look for the dynamic equation for $\hat{\rho}(t) \equiv \text{Tr}_B(\hat{\rho}_T(t))$, we trace out the bath degrees of freedom in the above equation and get

$$\frac{d\hat{\rho}}{dt} = -\alpha^2 \int_0^t ds \text{Tr}_B[\hat{H}_I(t), [\hat{H}_I(s), \hat{\rho}(t) \otimes \rho_B]], \quad (6)$$

where we have used the assumption $\rho_T(t) \approx \rho(t) \otimes \rho_B(0)$ and $\hat{\rho}_B(0) = \rho_B(0)$ denoted as ρ_B . The first term on the RHS disappears due to another requirement: $\text{Tr}_B[\hat{H}_I(t), \hat{\rho}_T(0)] = 0$. The physical meaning of it will be carried out later.

Born approximation*

Under Born approximation, the dynamics of $\hat{\rho}(t)$ is time-local but still non-Markovian because the integral from initial time $t = 0$ in the equation shows its evolution depends upon an explicit choice for the initial time.

The dynamics will become Markovian if we can push the lower limit of the integral to $-\infty$ without changing the result, which requires the integrand decay fast enough when s is away from t .

This requirement can be satisfied with the physical condition that specific correlation functions in the bath decay fast enough with time, which will be discussed later.

Jump operators

The Schrodinger picture interaction Hamiltonian is written as

$$H_I = \sum_{\alpha} S_{\alpha} \otimes B_{\alpha}. \quad (7)$$

(The Oxford book says S_{α} and B_{α} should be Hermitian, but it seems fine to go beyond it as long as H_I is Hermitian.)

To explicitly see how H_I affects the system's dynamics, we decompose S_{α} with the “eigenoperators” of H . An eigenoperator of H is defined as a **jump operators** connecting two different eigenstates of H . Therefore, by making use of the projection operator $P(E)$ onto each eigen-subspace of H with energy E , we have

$$S_{\alpha}(\omega) = \sum_{E' - E = \omega} P(E) S_{\alpha} P(E'), \quad (8)$$

where the summation extends over all eigen energies E, E' s.t. $E' - E = \omega$.

Jump operators

It is straightforward to verify that the $S_\alpha(\omega)$ operators sum up to S_α :

$$\sum_{\omega} S_\alpha(\omega) = \sum_{E, E'} P(E) S_\alpha P(E') = \left(\sum_E P(E) \right) S_\alpha \left(\sum_{E'} P(E') \right) = S_\alpha$$

The jump operators satisfy

$$[H, S_\alpha(\omega)] = -\omega S_\alpha(\omega), \quad [H, S_\alpha^\dagger(\omega)] = \omega S_\alpha^\dagger(\omega), \quad (9)$$

so they take simple forms in the interaction picture

$$\hat{S}_\alpha(\omega, t) \equiv e^{iHt} S_\alpha(\omega) e^{-iHt} = e^{-i\omega t} S_\alpha(\omega). \quad (10)$$

The interaction picture interaction Hamiltonian is then

$$\hat{H}_I(t) = \sum_{\alpha, \omega} e^{-i\omega t} S_\alpha(\omega) \otimes \hat{B}_\alpha(t) = \sum_{\alpha, \omega} e^{+i\omega t} S_\alpha^\dagger(\omega) \otimes \hat{B}_\alpha^\dagger(t), \quad (11)$$

where $\hat{B}_\alpha(t) \equiv e^{iH_B t} B_\alpha e^{-iH_B t}$.

Bath requirement 1*

We now turn to investigate the two requirements regarding the bath which we left before.

The first one is $\text{Tr}_B[\hat{H}_I(t), \hat{\rho}_T(0)] = 0$. Writing it out explicitly:

$$\text{Tr}_B[\hat{H}_I(t), \hat{\rho}_T(0)] = \sum_{\alpha} \hat{S}_{\alpha}(t) \hat{\rho}(0) \text{Tr}_B [\hat{B}_{\alpha}(t) \rho_B] - \hat{\rho}(0) \hat{S}_{\alpha}(t) \text{Tr}_B [\rho_B \hat{B}_{\alpha}(t)] \quad (12)$$

we see that the requirement can be satisfied by $\text{Tr}_B [\rho_B \hat{B}_{\alpha}(t)] = 0$, which is equivalent to $\langle B_{\alpha} \rangle = \text{Tr}_B [\rho_B B_{\alpha}] = 0$, i.e., the expectation value of operator B_{α} in bath is zero.

Bath requirements 2* - Markov approximation

The other requirement for the bath is to make it possible to extend the lower limit of the integral in the dynamic equation to $-\infty$, i.e., the integrand decays fast enough as s goes away from t . Expanding $\hat{H}_I(t)$ using (11), and change the variable to $u = t - s$, we have

$$\begin{aligned}\frac{d\hat{\rho}}{dt} &= -\alpha^2 \int_0^t du \text{Tr}_B[\hat{H}_I(t), [\hat{H}_I(t-u), \hat{\rho}(t) \otimes \rho_B]] \\ &= \alpha^2 \int_0^t du \text{Tr}_B[\hat{H}_I(t-u) \hat{\rho}(t) \otimes \rho_B \hat{H}_I(t) - \hat{H}_I(t) \hat{H}_I(t-u) \hat{\rho}(t) \otimes \rho_B] + \text{H.c.} \\ &= \sum_{\omega, \omega'} \sum_{\alpha, \beta} e^{i(\omega' - \omega)t} \Gamma_{\alpha\beta}(\omega) \left(S_{\beta}(\omega) \hat{\rho}(t) S_{\alpha}^{\dagger}(\omega') - S_{\alpha}^{\dagger}(\omega') S_{\beta}(\omega) \hat{\rho}(t) \right) + \text{H.c.}\end{aligned}\tag{13}$$

where the coefficient $\Gamma_{\alpha\beta}(\omega)$ is

$$\Gamma_{\alpha\beta}(\omega) \equiv \alpha^2 \int_0^t du e^{i\omega u} \langle \hat{B}_{\alpha}^{\dagger}(t) \hat{B}_{\beta}(t-u) \rangle.\tag{14}$$

Bath requirements 2* - Markov approximation

The requirement of extending the lower limit of $\int_0^t ds$ to $-\infty$ becomes extending the upper limit of $\int_0^t$ to ∞ , which can be satisfied if the bath correlation function (it is only a function of time difference u since $[H_B, \rho_B] = 0$)

$$\langle \hat{B}_\alpha^\dagger(t) \hat{B}_\beta(t-u) \rangle = \langle \hat{B}_\alpha^\dagger(u) \hat{B}_\beta(0) \rangle \quad (15)$$

is negligible after $u > t$, i.e., $\langle \hat{B}_\alpha^\dagger(u) \hat{B}_\beta(0) \rangle$ decays fast enough as u increases. It makes coefficients $\Gamma_{\alpha\beta}(\omega)$ the same for any time t , leading to a Markovian dynamic equation.

Note: A decay of the correlations can only be strictly valid for an infinitely large environment involving a continuum of frequencies. (See page 135 of the Oxford book for more details.)

Rotating wave approximation*

The equation we get now is usually called the Redfield equation, whose corresponding dynamical map is not guaranteed positive. To ensure complete positivity, We neglect the rapid oscillation terms in the previous equation derived, i.e., all the terms with $\omega' \neq \omega$

$$\frac{d\hat{\rho}}{dt} = \sum_{\omega} \sum_{\alpha, \beta} \Gamma_{\alpha\beta}(\omega) \left(S_{\beta}(\omega) \hat{\rho}(t) S_{\alpha}^{\dagger}(\omega) - S_{\alpha}^{\dagger}(\omega) S_{\beta}(\omega) \hat{\rho}(t) \right) + \text{H.c.} \quad (16)$$

To see the form of the Lindblad equation, we decompose the coefficients in a special way: $\Gamma_{\alpha\beta}(\omega) = \frac{1}{2}\gamma_{\alpha\beta}(\omega) + i\pi_{\alpha\beta}(\omega)$, such that

$$\begin{aligned} \pi_{\alpha\beta}(\omega) &= \frac{1}{2i}(\Gamma_{\alpha\beta}(\omega) - \Gamma_{\beta\alpha}^*(\omega)) \\ \gamma_{\alpha\beta}(\omega) &= \Gamma_{\alpha\beta}(\omega) + \Gamma_{\beta\alpha}^*(\omega) = \alpha^2 \int_{-\infty}^{\infty} du e^{i\omega u} \langle \hat{B}_{\alpha}^{\dagger}(u) \hat{B}_{\beta}(0) \rangle. \end{aligned} \quad (17)$$

The above two matrices for a fixed ω are Hermitian and the $\gamma(\omega)$ matrix can be proved positive (see Oxford book page 136).

Lamb shift

The resulting master equation in the interaction picture is then

$$\frac{d\hat{\rho}}{dt} = -i[H_{LS}, \hat{\rho}(t)] + \mathcal{D}[\hat{\rho}(t)], \quad (18)$$

where the Hermitian operator

$$H_{LS} \equiv \sum_{\omega} \sum_{\alpha\beta} \pi_{\alpha\beta}(\omega) S_{\alpha}^{\dagger}(\omega) S_{\beta}(\omega) \quad (19)$$

is called **Lamb shift** Hamiltonian, giving an extra unitary contribution to the dynamics. Note that $[H, H_{LS}] = 0$, the Lamb shift Hamiltonian only slightly changes the unperturbed energy levels of the system.

Note: Going back to the Schrodinger picture, the unitary part on the RHS of the dynamic equation becomes $-i[H + H_{LS}, \rho(t)]$.

Dissipation

The dissipation part of the dynamics is described by the dissipator:

$$\mathcal{D}[\cdot] = \sum_{\omega} \sum_{\alpha, \beta} \gamma_{\alpha\beta}(\omega) \left(S_{\beta}(\omega)(\cdot) S_{\alpha}^{\dagger}(\omega) - \frac{1}{2} \{ S_{\alpha}^{\dagger}(\omega) S_{\beta}(\omega), (\cdot) \} \right). \quad (20)$$

Going back to the Schrodinger picture, the form of $\mathcal{D}[\cdot]$ doesn't need to be changed, except for replacing $\hat{\rho}(t)$ with $\rho(t)$.

For each ω , one can diagonalize the matrix $\gamma(\omega)$ by a unitary transformation $U(\omega)$:

$$\gamma_j(\omega) \delta_{jk} = \sum_{\alpha\beta} U_{\alpha j}^*(\omega) \gamma_{\alpha\beta}(\omega) U_{\beta k}(\omega). \quad (21)$$

Transforming the jump operators accordingly, $L_j(\omega) = \sum_{\beta} U_{\beta j}^*(\omega) S_{\beta}(\omega)$ and $L_j^{\dagger}(\omega) = \sum_{\alpha} S_{\alpha}^{\dagger}(\omega) U_{\alpha j}(\omega)$, we show the dissipator has standard form

$$\mathcal{D}[\cdot] = \sum_{\omega} \sum_j \gamma_j(\omega) \left(L_j(\omega)(\cdot) L_j^{\dagger}(\omega) - \frac{1}{2} \{ L_j^{\dagger}(\omega) L_j(\omega), (\cdot) \} \right). \quad (22)$$

Impurity model jump operators

Back to our impurity model. The eigenstates and energies of H :

$$|0\rangle : E = 0, \quad f_+^\dagger |0\rangle \text{ and } f_-^\dagger |0\rangle : E = E_0, \quad f_+^\dagger f_-^\dagger |0\rangle : E = 2E_0 + U. \quad (23)$$

For simplicity, denote the eigenstates as $|0\rangle, |+\rangle, |-\rangle, |2\rangle$. Then the projection operators onto the eigen-subspaces are

$$P_0 = |0\rangle \langle 0|, \quad P_1 = |+\rangle \langle +| + |-\rangle \langle -|, \quad P_2 = |2\rangle \langle 2|. \quad (24)$$

The operator f_+ is thus decomposed into the following operators:

$$f_+(E_0) = P_0 f_+ P_1 = |0\rangle \langle +|, \quad f_+(E_0 + U) = P_1 f_+ P_2 = |-\rangle \langle 2|. \quad (25)$$

Similar for f_- :

$$f_-(E_0) = P_0 f_- P_1 = |0\rangle \langle -|, \quad f_-(E_0 + U) = P_1 f_- P_2 = -|+\rangle \langle 2| \quad (26)$$

Impurity model jump operators

In our model, we have four jump operators: f_+ , f_- , $f_+^\dagger$, $f_-^\dagger$, let's denote them as S_1, S_2, S_3, S_4 , and we know their decompositions:

$$\begin{aligned} S_1(E_0) &= |0\rangle \langle +|, & S_1(E_0 + U) &= |- \rangle \langle 2| \\ S_2(E_0) &= |0\rangle \langle -|, & S_2(E_0 + U) &= -|+ \rangle \langle 2| \\ S_3(-E_0) &= |+ \rangle \langle 0|, & S_3(-E_0 - U) &= |2 \rangle \langle -| \\ S_4(-E_0) &= |- \rangle \langle 0|, & S_4(-E_0 - U) &= -|2 \rangle \langle +|. \end{aligned} \tag{27}$$

They will be our jump operators in the Lindblad equation. The corresponding bath operators are (bath with continuous spectrum)

$$\begin{aligned} B_1 &= \int dk V_k^* c_{k+}^\dagger, & B_2 &= \int dk V_k^* c_{k-}^\dagger \\ B_3 &= \int dk V_k c_{k+}, & B_4 &= \int dk V_k c_{k-}. \end{aligned} \tag{28}$$

(The extra coefficient appears in converting from summation to integral: $\sum_k \rightarrow \frac{L}{2\pi} \int dk$ is already taken care of by properly normalizing V_k in the continuum k -space.)

Impurity model bath correlation functions

For the bath operators, we readily see that for the bath in an equilibrium state, i.e., a mixture of eigenstates of H_B , the only non-zero bath correlators are the ones with identical subscripts:

$$\langle \hat{B}_\alpha^\dagger(u) B_\alpha \rangle, \quad \alpha = 1, 2, 3, 4. \quad (29)$$

Evaluating them ($\alpha = 2, 4$ are likewise):

$$\begin{aligned} \langle \hat{B}_3^\dagger(u) B_3 \rangle &= \int dk dk' V_k^* V_{k'} e^{i\epsilon_k u} \langle c_{k+}^\dagger c_{k'+} \rangle \\ &= \int dk dk' V_k^* V_{k'} e^{i\epsilon_k u} \delta(k - k') \langle n_{k+} \rangle \\ &= \int dk |V_k|^2 e^{i\epsilon_k u} \langle n_{k+} \rangle \end{aligned} \quad (30)$$

$$\langle \hat{B}_1^\dagger(u) B_1 \rangle = \int dk |V_k|^2 e^{-i\epsilon_k u} \langle 1 - n_{k+} \rangle \quad (31)$$

Impurity model coupling coefficients (rough est.)

The coefficient matrix $\Gamma_{\alpha\beta}(\omega)$ is then already diagonal. Additionally, assume the bath correlator decay exponentially $\langle \hat{B}_\alpha^\dagger(u) B_\alpha \rangle \sim e^{-u/\tau_\alpha}$ (where $\tau_\alpha \omega \ll 1$ for ω being energy level differences in the system), the integral can be evaluated:

$$\Gamma_\alpha(\omega) \sim \alpha^2 \int_0^\infty du e^{i\omega u} e^{-u/\tau_\alpha} = \alpha^2 \frac{1}{\frac{1}{\tau_\alpha} - i\omega}, \quad (32)$$

thus the real and imaginary parts are

$$\begin{aligned} \pi_\alpha(\omega) &= \frac{1}{2i}(\Gamma_\alpha(\omega) - \Gamma_\alpha^*(\omega)) \sim \alpha^2 \frac{\omega}{\frac{1}{\tau_\alpha^2} + \omega^2}, \\ \gamma_\alpha(\omega) &= \Gamma_\alpha(\omega) + \Gamma_\alpha^*(\omega) \sim \alpha^2 \frac{\tau_\alpha}{1 + \omega^2 \tau_\alpha^2}. \end{aligned} \quad (33)$$

All these coefficients are small in comparison with the typical energy level difference in the system. Their relative magnitudes are determined mainly by the τ_α of the corresponding bath correlators.

Impurity model Lamb shift (rough est.)

The Lamb shift Hamiltonian is thus

$$\begin{aligned} H_{LS} &= \sum_{\omega} \sum_{\alpha} \pi_{\alpha}(\omega) S_{\alpha}^{\dagger}(\omega) S_{\alpha}(\omega) \\ &= (\pi_3(-E_0) + \pi_4(-E_0)) |0\rangle \langle 0| \\ &\quad + (\pi_1(E_0 + U) + \pi_2(E_0 + U)) |2\rangle \langle 2| \\ &\quad + (\pi_1(E_0) + \pi_4(-E_0 - U)) |+\rangle \langle +| \\ &\quad + (\pi_2(E_0) + \pi_3(-E_0 - U)) |-\rangle \langle -|. \end{aligned} \tag{34}$$

Due to the $SU(2)$ symmetry of the spin degree of freedom, $\pi_1(\omega) = \pi_2(\omega)$, $\pi_3(\omega) = \pi_4(\omega)$, then H_{LS} only slightly perturbs the energy levels (as $\omega\tau \ll 1$) without breaking the degeneracy.

We will put the Lamb shift effect aside for now.

Impurity model damping rates

In our model, we should assume bath operators $B_{1,2}$ ($B_{3,4}$) are annihilation (creation) operators of particles **on some local modes**. As the eigenmodes in the bath are plane waves in space, the $|V_k|$ function should be a wide peak in the k -space, and $|V_k| = |V_{-k}|$.

The damping rates $\gamma_\alpha(\omega)$ are Fourier transform of the bath correlators. For example, assuming the bath being 1D,

$$\begin{aligned}\gamma_3(\omega) &= \alpha^2 \int_{-\infty}^{\infty} du e^{i\omega u} \langle \hat{B}_3^\dagger(u) B_3 \rangle = \alpha^2 \int du \int_{-\pi}^{\pi} dk e^{i(\omega + \epsilon_k)u} |V_k|^2 \langle n_{k+} \rangle \\ &= 2\pi\alpha^2 \int_{-\pi}^{\pi} dk \delta(\omega + \epsilon_k) |V_k|^2 \langle n_{k+} \rangle = 4\pi\alpha^2 \int_0^{\pi} dk \dots \\ &= 4\pi\alpha^2 \int \frac{d\epsilon_k}{v_g(\epsilon_k)} \delta(\omega + \epsilon_k) |V_k|^2 \langle n_{k+} \rangle = 4\pi\alpha^2 \frac{|V_{k(-\omega)}|^2}{v_g(-\omega)} \langle n_{k(-\omega)+} \rangle, \end{aligned} \tag{35}$$

where $k(-\omega)$ denotes the k s.t. $\epsilon_k = -\omega$, and $v_g \equiv |d\omega/dk|$ is the group velocity at energy ω . We've assumed ω to be within the bath energy band.

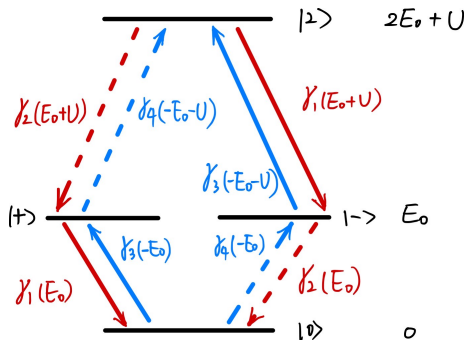
Impurity model damping rates

If ω is outside the energy band, the system, then the corresponding $\gamma_\alpha(\omega)$ vanishes due to the delta function. Assuming ω within bath energy band, the damping rates and their corresponding H_I terms are listed below:

$$\begin{aligned} f_+ \otimes \int dk V_k^* c_{k+}^\dagger : \quad \gamma_1(\omega) &= 4\pi\alpha^2 \frac{|V_{k(\omega)}|^2}{v_g(\omega)} \langle 1 - n_{k(\omega)+} \rangle \\ f_- \otimes \int dk V_k^* c_{k-}^\dagger : \quad \gamma_2(\omega) &= 4\pi\alpha^2 \frac{|V_{k(\omega)}|^2}{v_g(\omega)} \langle 1 - n_{k(\omega)-} \rangle \\ f_+^\dagger \otimes \int dk V_k c_{k+} : \quad \gamma_3(\omega) &= 4\pi\alpha^2 \frac{|V_{k(-\omega)}|^2}{v_g(-\omega)} \langle n_{k(-\omega)+} \rangle \\ f_-^\dagger \otimes \int dk V_k c_{k-} : \quad \gamma_4(\omega) &= 4\pi\alpha^2 \frac{|V_{k(-\omega)}|^2}{v_g(-\omega)} \langle n_{k(-\omega)-} \rangle \end{aligned} \tag{36}$$

Impurity model dissipator

It is instructive to show the damping rates and the corresponding jump operators using the following energy level diagram of the system:



$$\gamma_1(\omega) = 4\pi\alpha^2 \frac{|V_k(\omega)|^2}{v_g(\omega)} \langle 1 - n_{k(\omega)+} \rangle$$

$$\gamma_2(\omega) = 4\pi\alpha^2 \frac{|V_k(\omega)|^2}{v_g(\omega)} \langle 1 - n_{k(\omega)-} \rangle$$

$$\gamma_3(\omega) = 4\pi\alpha^2 \frac{|V_k(-\omega)|^2}{v_g(-\omega)} \langle n_{k(-\omega)+} \rangle$$

$$\gamma_4(\omega) = 4\pi\alpha^2 \frac{|V_k(-\omega)|^2}{v_g(-\omega)} \langle n_{k(-\omega)-} \rangle$$

Impurity model dissipator

The dissipator for the impurity model is a super operator:

$$\mathcal{D}[\cdot] = \sum_{\omega} \sum_j \gamma_j(\omega) \left(S_j(\omega)(\cdot) S_j^{\dagger}(\omega) - \frac{1}{2} \{ S_j^{\dagger}(\omega) S_j(\omega), (\cdot) \} \right). \quad (37)$$

In the double basis $|a\rangle \langle b|$, it takes a matrix form:

$$D_{ab,cd} \equiv \text{Tr}(|b\rangle \langle a| \mathcal{D}[|c\rangle \langle d|]) \quad (38)$$

These matrix elements are

$$\begin{aligned} D_{ab,ab} &= -\frac{1}{2} \sum_c \gamma_{bc} + \gamma_{ca}, \\ D_{aa,bb} &= \gamma_{ab} \quad (a \neq b), \end{aligned} \quad (39)$$

where γ_{ab} is the damping rate $\gamma_j(E_b - E_a)$ associated with the jump operator $S_j(E_b - E_a) = |a\rangle \langle b|$.

Thermal equilibrium: KMS relation

We would like to ask: if the bath is in thermal equilibrium (Gibbs state) $\rho_B = e^{-\beta(H_B - \mu N_B)} / Z_B$, will the thermal state $\rho_{\text{th}} = e^{-\beta(H - \mu N)} / Z$ be a steady state of the system given by the Lindblad equation?

As $[H, \rho_{\text{th}}] = [H_{LS}, \rho_{\text{th}}] = 0$, we only need to prove $\mathcal{D}[\rho_{\text{th}}] = 0$. For bath being in thermal state, the bath correlators satisfy Kubo–Martin–Schwinger (KMS) condition:

$$\langle \tilde{B}_\alpha^\dagger(t) B_\beta \rangle = \langle B_\beta \tilde{B}_\alpha^\dagger(t + i\beta) \rangle, \quad (40)$$

where $\tilde{O}_B^\dagger(\tau) \equiv e^{i(-\mu N_B)\tau} \hat{O}_B(\tau) e^{-i(-\mu N_B)\tau}$ generalizes Heisenberg picture.

As B_α are single particle creation/annihilation operators, $[N_B, B_\alpha^\dagger] = \pm B_\alpha^\dagger$ (+ for B_α being annihilation). Then $\tilde{B}_\alpha^\dagger(\tau) = e^{\mp i\mu\tau} \hat{B}_\alpha^\dagger(\tau)$, and thus the KMS condition for our case becomes

$$\langle \hat{B}_\alpha^\dagger(t) B_\beta \rangle = e^{\pm \beta\mu} \langle B_\beta \hat{B}_\alpha^\dagger(t + i\beta) \rangle, \quad (41)$$

a modified version of the KMS condition on page 137 of the Oxford book.

Thermal equilibrium: damping rates relation

For our model, the KMS conditions are

$$\begin{aligned}\langle \hat{B}_{1,2}^\dagger(t) B_{1,2} \rangle &= e^{-\beta\mu} \langle B_{1,2} \hat{B}_{1,2}^\dagger(t + i\beta) \rangle = e^{-\beta\mu} \langle B_{3,4}^\dagger \hat{B}_{3,4}(t + i\beta) \rangle, \\ \langle \hat{B}_{3,4}^\dagger(t) B_{3,4} \rangle &= e^{\beta\mu} \langle B_{3,4} \hat{B}_{3,4}^\dagger(t + i\beta) \rangle = e^{\beta\mu} \langle B_{1,2}^\dagger \hat{B}_{1,2}(t + i\beta) \rangle.\end{aligned}\quad (42)$$

Taking Fourier transform, we have:

$$\begin{aligned}\gamma_{1,2}(\omega) &= \int_{-\infty}^{\infty} du e^{i\omega u} \langle \hat{B}_{1,2}^\dagger(u) B_{1,2} \rangle = e^{-\beta\mu} \int_{-\infty}^{\infty} du e^{i\omega u} \langle B_{3,4}^\dagger \hat{B}_{3,4}(u + i\beta) \rangle \\ &= e^{-\beta\mu} \int_{-\infty}^{\infty} du e^{i\omega u} \langle \hat{B}_{3,4}^\dagger(-u - i\beta) B_{3,4} \rangle \\ (\tau = -u - i\beta) &= e^{-\beta\mu} e^{\beta\omega} \int_{-\infty - i\beta}^{\infty - i\beta} d\tau e^{-i\omega\tau} \langle \hat{B}_{3,4}^\dagger(\tau) B_{3,4} \rangle = e^{\beta(\omega - \mu)} \gamma_{3,4}(-\omega),\end{aligned}\quad (43)$$

where we have assumed $\langle \hat{B}_{3,4}^\dagger(\tau) B_{3,4} \rangle$ being analytic in the stripe from real axis ($\text{Im}(\tau) = 0$) to $\text{Im}(\tau) = -i\beta$.

Thermal equilibrium: damping rates relation

The relation between the two sets of damping rates we got from the KMS condition is

$$\gamma_{1,2}(\omega) = e^{\beta(\omega-\mu)} \gamma_{3,4}(-\omega), \quad (44)$$

which is identical to what we had before for the 1D bath:

$$\frac{\gamma_{1,2}(\omega)}{\gamma_{3,4}(-\omega)} = \frac{1 - n_{\beta,\mu}(\omega)}{n_{\beta,\mu}(\omega)} \quad (45)$$

Using the KMS condition, it is obvious that when the free fermion bath is in its thermal equilibrium, the ratio between the damping rates of one dissipation process (jumping from one system eigenstate to another) and its reverse process is fully determined by the occupation number $n_{\beta,\mu}(\omega)$ on the corresponding bath energy level ω . This occupation number $n_{\beta,\mu}(\omega)$, given by the Fermi-Dirac distribution, is a function of β and μ solely, independent of other details of the bath (unless the energy ω is in a band gap, then the rates are zero).

Thermal equilibrium of the impurity model

To show $\mathcal{D}[\rho_{\text{th}}] = 0$, the last piece of the puzzle is found by the following relation:

$$\rho_{\text{th}} S_{\alpha}(\omega) = e^{\beta(\omega - \mu \Delta N)} S_{\alpha}(\omega) \rho_{\text{th}}, \quad (46)$$

where $\Delta N = N_b - N_a$ is the particle number difference between the two eigenstates connected by $S_{\alpha}(\omega) = |a\rangle \langle b|$, just like $\omega = E_b - E_a$. We have

$$\begin{aligned} S_{\alpha}(\omega) \rho_{\text{th}} S_{\alpha}^{\dagger}(\omega) &= e^{-\beta(\omega - \mu \Delta N)} S_{\alpha}(\omega) S_{\alpha}^{\dagger}(\omega) \rho_{\text{th}}, \\ \rho_{\text{th}} S_{\alpha}^{\dagger}(\omega) S_{\alpha}(\omega) &= S_{\alpha}^{\dagger}(\omega) S_{\alpha}(\omega) \rho_{\text{th}}. \end{aligned} \quad (47)$$

As for the impurity model, $S_{1,2}^{\dagger}(\omega) = S_{3,4}(-\omega)$ and $\gamma_{1,2}(\omega) = e^{\beta(\omega - \mu)} \gamma_{3,4}(-\omega)$, it is straightforward to show that the terms involving $S_{1,2}(\omega)$ cancel with those involving $S_{3,4}(-\omega)$, thus $\mathcal{D}[\rho_{\text{th}}] = 0$, leading to

$$\mathcal{L}[\rho_{\text{th}}] = 0. \quad (48)$$

Discussion

In the Oxford textbook, S_α and B_α in H_I are required to be Hermitian operators. We go beyond this assumption to include particle exchanging processes between the system and bath. By generalizing the proof on page 137 of the textbook, we show that the thermal equilibrium state $\rho_{\text{th}} = e^{-\beta(H-\mu N)} / Z$ from the grand canonical ensemble is a steady state of the Lindblad equation, in place of the thermal state from the canonical ensemble.

Although we proved ρ_{th} is a steady state, it might not be the only steady state of the system, especially when there is degeneracy in the spectrum of H .

Even if we do have a system with a unique steady state, it is still not true that for any arbitrary initial state, the system relaxes to equilibrium. The concept of ergodicity is needed to deal with this part of the problem.